Estimated computational cost and API pricing

Frozen two-stage DEMA pipeline and sensitivity analysis

Pricing assumptions

Costs were estimated in US dollars using an illustrative GPT-4.1 reference rate of $2.00 per million input tokens and $8.00 per million output tokens. The calculation is: cost = (input tokens / 1,000,000 × $2.00) + (output tokens / 1,000,000 × $8.00). These are reference list rates rather than a reconstruction of the institution's Azure invoice. Azure region, service tier, negotiated terms, student credits and prompt-caching discounts may change the amount actually charged. Current contractual rates should therefore be verified in the Azure portal. Public reference pages: https://openai.com/api/pricing/ and https://azure.microsoft.com/pricing/details/cognitive-services/openai-service/.

Observed token use

Component	Calls	Input tokens	Output tokens	Estimated cost
Stage 1: 30 held-out chunks	30	329,464	60,708	$1.14
Stage 2: validated final assessment	16	3,674,118	41,323	$7.68
Validated final-pipeline execution	46	4,003,582	102,031	$8.82
Two-indicator sensitivity experiment	6	1,190,439	12,873	$2.48

Interpretation of the final-pipeline estimate

The validated final-pipeline execution used 4,003,582 input tokens and 102,031 output tokens, corresponding to approximately $8.82 at the assumed rates. The final pipeline is defined here as 46 validated calls: 30 frozen Stage 1 calls for the held-out chunks and 16 Stage 2 calls, including the recorded schema retry. Discarded and interrupted calls are excluded. The recently completed two-indicator experiment added six calls and approximately $2.48.

Estimated prompt-development cost

Development activity	Estimated calls	Estimated input	Estimated output	Estimated cost
Five Stage 1 iterations (P1-P5)	150	1,703,470	510,315	$7.49
Five Stage 2 iterations (P1-P5)	80	11,900,720	206,615	$25.45
Combined prompt iterations	230	13,604,190	716,930	$32.94

The prompt-development estimates treat the five requested iterations as the P1-P5 refinement cycles beyond P0. For Stage 1, each iteration is assumed to process all 30 development chunks. Input was calculated from the actual development files; output was estimated from the observed token yield per accepted evidence record. For Stage 2, each iteration is conservatively treated as one complete 16-call scoring pass over the 390-record development bank. Synthetic boundary-case calls are not included, so the estimates should be interpreted as lower-bound planning values.

Estimated full-corpus pipeline run

Component	Estimated calls	Estimated input	Estimated output	Estimated cost
Stage 1: all 60 processing chunks	60	672,933	162,771	$2.65
Stage 2: complete evidence bank	16	3,674,118	41,323	$7.68
Full pipeline from source chunks	76	4,347,051	204,094	$10.33

The directory final_substantive_corpus_files contains 60 Stage 1 processing units: 30 held-out chunks under chunks and 30 development chunks under development sources. The 14 files under converted are source-level representations and were not counted as additional extraction units. A complete from-source execution is therefore estimated at 76 calls: 60 Stage 1 calls plus the same 16-call Stage 2 procedure used for the validated assessment. The projected cost is approximately $10.33.